



AI Playbook and Toolkit for Public Health Departments

Respecting Tribal Sovereignty and Indigenous Data Governance in Public Health AI Projects

GOV 110

This module introduces public health staff to the responsibilities that arise when AI projects involve Tribal Nations, American Indian and Alaska Native participants, or health data that may describe Tribal citizens, communities, lands, services, or priorities. The central message is that Tribal Nations are sovereign governments, not stakeholder groups, and their health data cannot be treated as ordinary administrative information. AI projects must be shaped through early consultation, formal partnership, and clear agreements before data are collected, transformed, analyzed, shared, or used to train, test, or deploy AI systems. This module is intended for programmatic, technical, analytic, governance, policy, communications, and leadership staff who may participate in AI planning or implementation. The module emphasizes Indigenous Data Sovereignty, Indigenous Data Governance, Digital Sovereignty, and cultural humility as core public health AI competencies. Learners will examine how AI project decisions can affect Tribal authority, community trust, privacy, group harm, health equity, interpretation of findings, and control over future data use. The module also addresses practical safeguards for working with AI tools, including large language models, automated transcription, data linkage, geospatial analysis, predictive models, and external platforms. The goal is not to make every learner an expert in Tribal law, but to help staff recognize when Tribal sovereignty is implicated and when formal Tribal engagement, legal review, and governance review are required.

Level	Introductory
Primary track	Governance and Security
Appears in tracks	governance-security, shared-foundational, policy, program-management

Learning Objectives

- Explain why Tribal sovereignty must be considered before starting AI projects that involve Tribal Nations, American Indian and Alaska Native participant data, or health data that may be connected to Tribal citizens, communities, lands, services, or priorities.
- Describe Indigenous Data Sovereignty, Indigenous Data Governance, Digital Sovereignty, and the CARE Principles in plain language, and explain how these concepts apply to AI use case intake, data management, model development, evaluation, dissemination, and lifecycle monitoring.
- Identify governance questions that should be addressed before using Tribal data or AI tools in public health projects, including questions about Tribal consultation, Tribal laws and policies, Tribal IRBs or research review processes, data sharing agreements, publication review, access controls, and permitted or prohibited uses.
- Distinguish cultural competence from cultural humility and explain why Tribal-specific consultation, relationship building, and respect for government-to-government authority are required rather than relying on generic cultural awareness or one-size-fits-all engagement practices.
- Identify AI project risks involving inferred Tribal affiliation, inappropriate geographic inference, external platform exposure, model outputs that stigmatize communities, dissemination that bypasses Tribal review, and secondary uses that were not approved by Tribal partners.
- Connect sovereignty protections to practical project artifacts, including Tribal data sharing agreements, Data Management and Sharing plans, review memos, community benefit statements, AI system documentation, dissemination plans, and monitoring or sunset criteria.
- Draft an initial Tribal sovereignty and Indigenous data governance screening memo for a proposed AI-supported public health workflow that identifies whether the use case should proceed, pause for consultation, be revised, or move to formal Tribal, legal, privacy, security, equity, and governance review.

Module Overview

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Definitions

- **Tribal sovereignty:** The inherent authority of Tribal Nations to govern themselves and protect the health, safety, welfare, and future of their citizens.

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- **Indigenous Data Sovereignty:** The right of Indigenous Peoples and Nations to govern data about them, their citizens, lands, resources, knowledge, and communities.
 - **Indigenous Data Governance:** The policies, agreements, relationships, roles, and oversight mechanisms that put Indigenous Data Sovereignty into practice.
 - **CARE Principles:** A framework for Indigenous Data Governance organized around Collective Benefit, Authority to Control, Responsibility, and Ethics.
 - **Digital Sovereignty:** Control over digital environments, platforms, infrastructure, and technical systems that store, process, move, or expose data.
 - **Cultural humility:** An ongoing practice of learning, accountability, and respect that recognizes power imbalances and Tribal-specific protocols.

Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace Tribal law, Tribal policy, Tribal consultation, Tribal IRB review, legal counsel, privacy review, security review, civil rights review, procurement review, or agency policy review. Requirements may vary by Tribal Nation, jurisdiction, funding source, data sharing agreement, research status, operational context, and data type. Learners should confirm applicable requirements before approving, procuring, deploying, sharing, or publicly communicating about AI-supported workflows involving Tribal Nations or AI/AN participant data.

Why This Topic Matters for Public Health AI

AI projects can magnify the consequences of weak data governance because AI systems depend on data extraction, transformation, linkage, model training, inference, and reuse. When health data involve Tribal Nations or AI/AN participants, those consequences can include individual privacy harms, community stigmatization, misclassification, loss of control over data, misinterpretation of culturally situated information, or use of data for purposes that were never agreed to. Public health agencies may intend to improve services, reduce disparities, or strengthen emergency response, but good intent does not replace Tribal authority to govern data and research involving Tribal citizens, communities, and jurisdictions. Respecting sovereignty at the start of an AI project prevents downstream harm that may be difficult or impossible to repair.

This topic also matters because many AI workflows blur the boundary between research, operations, analytics, and communications. A project that begins as an internal dashboard, chatbot, transcription workflow, or model evaluation may later influence public messaging, resource allocation, surveillance interpretation, or program decisions. If Tribal partners are consulted only after design choices are made, they may have little opportunity to shape the data sources, consent terms, model outputs, storage location, access controls, or review procedures. Meaningful partnership requires engagement while the project can still change.

Public health AI work also occurs within a history of research abuse, data misuse, underinvestment, and misrepresentation affecting AI/AN communities. NIH guidance on responsible management and sharing of AI/AN participant data emphasizes that biomedical research involving Tribal Nations must be predicated on respect for Tribal sovereignty and awareness that failures to honor sovereignty have caused stigmatization and other harms. The CARE Principles similarly emphasize that data governance must address collective benefit, authority to control, responsibility, and ethics rather than focusing only on technical openness or reuse. For AI projects, these principles require teams to ask who benefits, who controls the data, who is accountable, and whether the work aligns with Tribal laws, policies, preferences, and values.

Public Health Example

A state health department wants to develop an AI-assisted analytics workflow to summarize emergency department syndromic surveillance notes and identify patterns related to respiratory illness. The initial technical design includes free-text chief complaint data, facility location, ZIP code, race and ethnicity fields, and model-generated summaries by county. Several counties include Tribal lands, and some small-area summaries could indirectly identify Tribal communities or lead staff to infer Tribal affiliation from residence, facility use, or geography. The project team initially views the work as routine surveillance improvement, but the intake review identifies that Tribal sovereignty and Indigenous data governance must be addressed before development proceeds.

A sovereignty-centered approach would pause the technical build and begin with appropriate Tribal consultation and partnership discussions. The team would identify which Tribal Nations may be affected, whether Tribal public health authorities already receive or govern relevant data, whether Tribal laws or review processes apply, and what limitations should be placed on geographic analysis, small-number reporting, public dissemination, and AI-generated interpretation. The project would not use external AI platforms to process identifiable or potentially sensitive data unless the environment, data use, retention terms, and review pathway were approved. The team would also document whether the project creates community benefit defined by Tribal partners, not only operational benefit for the state agency.

The final project might look different from the original proposal. Tribal partners could request a Tribal-specific data sharing agreement, a joint review group, suppression rules for small numbers, limits on geospatial inference, local review before public summaries, a requirement to return results in a useful format, or a decision that some data should not be used for AI modeling. The project may also identify capacity-building opportunities, such as supporting Tribal epidemiology staff, providing access to relevant outputs, or co-developing monitoring measures. The key lesson is that sovereignty review does not merely add a compliance step; it can reshape the purpose, design, governance, and success measures of the AI project.

Technical, Programmatic, and Operational Deep Dive

AI project teams should begin by asking whether the proposed use involves Tribal Nations, Tribal citizens, AI/AN participants, Tribal lands, Tribal service settings, Tribal public health authorities, or data elements that could be used to infer Tribal affiliation. This question should be asked during use case intake, not after a data set has been assembled. The answer may not be obvious because Tribal relevance can appear through geography, facility relationships, program participation, small-area reporting, environmental exposure data, biospecimens, genetic information, language data, or community narratives. If the answer is uncertain, the team should treat the project as requiring additional review rather than assuming no Tribal implications exist.

Early consultation should occur before AI development, procurement, vendor demonstrations, pilot testing, or publication planning. Consultation is not the same as sending a completed plan for comment. It should allow Tribal Nations to influence the project purpose, data scope, roles, benefits, risks, consent expectations, storage location, review process, and dissemination strategy. Timelines and budgets should recognize that government-to-government consultation, community review, and relationship building require time and resources.

Formal agreements are essential because AI projects create many opportunities for data reuse and secondary interpretation. Agreements may need to address data ownership or stewardship, permitted uses, prohibited uses, data retention, destruction or return, access rights, cloud or vendor environments, model training restrictions, publication review, benefit sharing, incident response, and conditions for future use. Where NIH-funded research is involved, the Data Management and Sharing Plan should incorporate AI/AN data management and sharing practices and Tribal preferences. For operational public health projects outside NIH research, similar principles should still guide data sharing agreements, memoranda of understanding, and governance records.

AI teams should not infer Tribal affiliation from residential location, ZIP code, facility use, names, language, or other proxy variables unless that inference has been explicitly approved and is necessary for a mutually agreed public health purpose. Proxy inference can create privacy risk, misclassification, and community harm. It can also

undermine Tribal authority if outside organizations create AI-derived labels about Tribal identity or community membership without consent. If affiliation or Tribal relevance is needed, the project should address how that information is defined, who controls it, whether it should be collected, and whether the Tribal Nation approves the method.

Data management controls should be designed before data are extracted or transformed for AI use. Teams should classify data sensitivity, separate identifiable data from analytic data when appropriate, minimize variables, restrict access, log use, and define approved environments. External AI tools, automated transcription services, large language models, coding assistants, and cloud platforms should not receive Tribal data, identifiable information, sensitive community information, or non-public operational details unless the tool and environment have been reviewed and approved. Contract terms should address retention, vendor access, model training, subcontractors, data location, deletion, incident notification, and audit rights.

Model development and evaluation should include sovereignty-aware validation questions. The team should ask whether training data represent the community appropriately, whether small sample sizes make results unreliable, whether model outputs could stigmatize a Tribal Nation, and whether performance should be reported separately or suppressed. The team should also ask who has authority to interpret findings and what contextual knowledge is needed to avoid harmful conclusions. Evaluation should include human review by people with appropriate community, program, epidemiologic, technical, and governance expertise.

Communication and dissemination require special care because AI outputs can be polished, confident, and wrong. Public-facing summaries, dashboards, maps, chatbots, and reports may create harm if they identify small communities, imply causation without evidence, frame disparities as community deficits, or omit historical and structural context. Tribal partners may require pre-review of reports, presentations, manuscripts, dashboard text, public advisories, or model documentation. Teams should document dissemination approvals and avoid treating public release as a default endpoint.

Cultural humility should be built into the project operating model. Staff should learn the history, governance structure, communication protocols, and preferred engagement processes of the Tribal Nation they are working with, while recognizing that this knowledge does not make them authorities on the community. Meetings should be scheduled and facilitated in ways that respect Tribal leadership, community priorities, and local decision-making processes. Compensation, authorship, data access, capacity building, and benefit sharing should reflect the value of Tribal expertise and partnership.

Risks, Failure Modes, and Guardrails

A major risk is treating Tribal health data as if it were ordinary administrative or research data controlled solely by the institution that currently holds it. This can lead to AI development without Tribal consultation, data sharing without appropriate authority, or model outputs that affect Tribal communities without Tribal review. The failure mode is often procedural rather than malicious: staff follow routine analytics practices without recognizing that sovereignty changes the decision pathway. The guardrail is to require Tribal sovereignty screening during AI use case intake and to route potentially relevant projects for legal, privacy, governance, and Tribal consultation review before data are used.

Another risk is using external AI platforms for transcription, summarization, coding, translation, or drafting with sensitive Tribal data or community information. These tools may retain prompts, store uploaded files, expose information to vendors or subcontractors, or use content in ways that are not transparent to the user. Even de-identified text can contain place names, small population details, cultural references, or contextual clues that create re-identification or group harm. The guardrail is to prohibit use of non-approved external AI tools with Tribal data and to require approved environments, contract review, and data minimization before any AI processing occurs.

A third risk is allowing model outputs to create or reinforce harmful narratives about Tribal communities. Predictive models, cluster analyses, risk scores, geospatial maps, or AI-generated summaries may overemphasize deficits, ignore structural determinants, misinterpret small numbers, or produce misleading comparisons. These outputs can

affect public trust, funding decisions, program priorities, or media narratives. The guardrail is to require sovereignty-aware review of model purpose, variables, outputs, interpretation, dissemination, and monitoring before results are shared or used operationally.

A fourth risk is considering equity, accessibility, and community benefit too late. If the project is already funded, contracted, designed, or near deployment before Tribal partners are engaged, meaningful changes may be difficult. This can create extractive relationships in which data and expertise flow outward while benefits, authority, and infrastructure do not flow back to the community. The guardrail is to define community benefit, capacity building, shared decision-making, and return-of-results expectations at the beginning of the project.

A fifth risk is assuming that one Tribal engagement process applies to all Tribal Nations. Each Tribal Nation has its own government, history, legal requirements, research review expectations, cultural protocols, and data priorities. A process accepted by one Tribal Nation cannot be generalized to another without permission. The guardrail is to identify each affected Tribal Nation and confirm the appropriate consultation, approval, and documentation pathway for each one.

A sixth risk is failing to maintain accountability after deployment. AI systems may change as data pipelines, vendors, model versions, workflows, community conditions, or public health priorities change. Without lifecycle governance, a use that was conditionally approved may drift beyond the original agreement. The guardrail is to maintain an inventory of AI uses involving Tribal data, monitor compliance with approval conditions, review incidents, revisit agreements when scope changes, and retire or pause systems when they no longer align with Tribal preferences or public health benefit.

Application to AI-Supported Workflows

For AI use case intake, teams should add screening questions that identify whether the proposed project involves Tribal Nations, AI/AN participant data, Tribal lands, Tribal service settings, Tribal public health authorities, small-area geography, biospecimens, genetic data, environmental exposure data, or other information that could affect Tribal communities. The intake form should require the proposer to identify known Tribal partners, anticipated benefits, possible harms, data sources, external tools, and whether consultation has occurred. A project should not advance to procurement, model development, pilot testing, or public release until the sovereignty screening has been resolved. This makes Tribal sovereignty a front-end design requirement rather than a late-stage approval issue.

For data preparation and model development, teams should document permitted data uses and technical restrictions before creating analytic files. This includes rules for variable selection, de-identification, linkage, geospatial analysis, small number suppression, feature engineering, model training, validation, storage, access, and deletion. Teams should also decide whether data can be used in model training, whether outputs can be reused in future projects, and whether a model trained with Tribal data can be transferred to other jurisdictions or use cases. These decisions should be made through formal agreement, not by default technical convenience.

For deployment and monitoring, teams should define who reviews outputs, who receives results, who can change the system, and who can decide that the system should pause or stop. Monitoring should include accuracy, fairness, usability, community impact, privacy incidents, scope creep, and compliance with Tribal agreements. If the AI tool supports public communication, resource prioritization, surveillance interpretation, or program decisions, Tribal partners should have a clear pathway to question or challenge outputs. The system should be designed to preserve Tribal authority throughout the lifecycle, not only during initial approval.

Reflection Questions

Before using AI with health data that may involve Tribal Nations or AI/AN participants, learners should pause and consider whether they know who has authority to approve the use. They should ask whether the project has identified affected Tribal Nations, relevant Tribal laws or review processes, and the appropriate form of consultation or partnership. They should also consider whether the proposed AI use creates community benefit as defined by

Tribal partners, rather than only efficiency for an external agency. These questions help teams recognize that sovereignty is a design condition, not a courtesy step.

Learners should also reflect on how the project could create harm even if no individual is directly identified. Could the AI output stigmatize a community, reveal small-area patterns, infer Tribal affiliation, misinterpret cultural context, or support decisions that affect services or resources? Could an external AI tool retain or reuse information that the project team does not have authority to share? These questions should be answered before development begins, when the project can still be changed.

Practical Exercise

Select a proposed AI-supported public health project that could involve Tribal Nations, AI/AN participant data, Tribal lands, Tribal service settings, or data that could indirectly identify or affect Tribal communities. The project may be a surveillance summary tool, predictive model, automated transcription workflow, chatbot, public communication assistant, environmental health mapping project, data linkage project, or research analysis. Complete a Tribal sovereignty and Indigenous data governance screening memo for the project. The memo should identify the public health purpose, data sources, affected communities, potential Tribal authority concerns, consultation needs, data management safeguards, external platform risks, review bodies, and recommended next steps.

Example response: A state agency proposes to use an AI tool to summarize open-text case investigation notes for emerging respiratory illness trends. The initial data include case narratives, county, ZIP code, facility, race and ethnicity, and outbreak location fields, and several records may involve Tribal service areas or small communities near Tribal lands. The screening memo identifies that the project should not send free-text notes to an external AI platform, should not infer Tribal affiliation from geography, and should not produce small-area summaries until Tribal consultation and legal review occur. The recommended next step is to meet with potentially affected Tribal public health authorities to determine whether the project has a mutually agreed purpose, what data may be used, what outputs are useful, what findings require pre-review, and what safeguards should be documented in a data sharing agreement.

Expected Artifact or Evidence

The expected artifact is a Tribal Sovereignty and Indigenous Data Governance Screening Memo for one AI-supported public health use case. The memo should be complete enough for a supervisor, AI governance body, Tribal liaison, legal counsel, privacy officer, security lead, equity reviewer, or Tribal partner to understand why the project may implicate Tribal sovereignty. It should distinguish known facts from assumptions and identify what cannot be decided until appropriate Tribal consultation occurs. The memo should not include protected health information, personally identifiable information, or confidential Tribal information.

A strong memo includes the proposed AI use, public health purpose, data sources, affected Tribal Nations or uncertainty about Tribal relevance, AI tools or vendors under consideration, data storage and processing environment, anticipated outputs, potential benefits, potential harms, and recommended review pathway. It should address CARE Principles, data management expectations, cultural humility, consultation needs, and whether formal agreements are required before the project proceeds. The artifact may be used as an intake attachment, governance briefing, or planning document, but it does not replace Tribal approval or formal legal review.

Expected Assignment

- Tribal Sovereignty and Indigenous Data Governance Screening Memo

Knowledge Check

- 1. Why should Tribal sovereignty be considered before an AI project with health data begins?

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- 2. Which statement best reflects Indigenous Data Sovereignty?
 - 3. Which AI practice should be avoided unless explicitly approved through the appropriate governance pathway?
 - 4. What is the strongest reason to involve Tribal partners early in the AI lifecycle?
 - 5. What should a sovereignty-aware AI project do when Tribal relevance is uncertain?

References and Resources

- Global Indigenous Data Alliance. CARE Principles for Indigenous Data Governance.: <https://www.gida-global.org/care>
- National Institutes of Health. NOT-OD-22-214: Supplemental Information to the NIH Policy for Data Management and Sharing: Responsible Management and Sharing of American Indian/Alaska Native Participant Data.: <https://grants.nih.gov/grants/guide/notice-files/NOT-OD-22-214.html>
- National Institutes of Health Tribal Health Research Office. Tribal Health Research Office resources.: <https://dpcpsi.nih.gov/thro>
- United States Indigenous Data Sovereignty Network. Resources and information on Indigenous Data Sovereignty.: <https://usindigenousdata.org/>
- All of Us Research Program. Tribal engagement and data use resources.: <https://www.researchallofus.org/>
- HHS and CDC Tribal consultation resources. Use relevant agency consultation policies and Tribal liaison guidance before starting projects involving Tribal Nations or AI/AN data.
- Brookings. Digital sovereignty and Indigenous Nations resources.: <https://www.brookings.edu/>